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Date: 12/01/25

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Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 4

1. A student remarks: "It doesn't make sense that intermediate goods¹ are not counted in GDP². A cotton is an intermediate good, and without it a shirt won't be produced. So why don't we count the cotton in GDP?" Provide an answer for the student's question.
- ☐ A. There is no value added in the production of cotton. Therefore, it is not included in the value-added method of measuring GDP.
 - ☒ B. The value of the cotton is not directly counted in GDP, but it is included in the value-added method of measuring GDP.
 - ☐ C. The value of the cotton is related to transfer payments that are not included in GDP because they are not received in exchange for producing a new good or service.
 - ☐ D. The value of both final goods and intermediate goods is included in the calculation of GDP.

1: More Info

Intermediate good or service is a good or service that is an input into another good or service, such as a tire on a truck.

2: More Info

Gross domestic product (GDP) is the market value of all final goods and services produced in a country during a period of time, typically one year.

2. Indicate whether each of the following transactions represents the purchase of a final good³.
- a. The purchase of tires from Bridgestone tire company by an automobile manufacturer. This is not the purchase of a final good .
 - b. The purchase of an aircraft carrier by the federal government. This is the purchase of a final good .
 - c. The purchase of French wine by a Canadian consumer. This is the purchase of a final good .
 - d. The purchase of a new airliner by WestJet. This is the purchase of a final good .

3: More Info

Final good or service is a good or service purchased by a final user.

3. Indicate whether you agree or disagree with the following statement:
- "In years when people buy few shares of stock, investment will be low and, therefore, so will GDP⁴."
- Choose the correct answer below.

- ☐ A. Disagree: While GDP will be low in this case, it is the result of a decrease in consumer expenditure on stocks, not investment spending.
- ☐ B. Agree: When investment is low, people must have less money to spend. Therefore, GDP decreases.
- ☐ C. Agree: $GDP = C + I + G + NX$. Therefore, as "I" (Investment) decreases, GDP decreases.
- ☒ D. Disagree: Investment as a component of GDP refers to the purchase of physical and human capital and inventory, not stock purchases.

4: More Info

Gross domestic product (GDP) is the market value of all final goods and services produced in a country during a period of time, typically one year.

4. Indicate whether each of the following is a final good, an intermediate good, or neither.

Coffee beans purchased by a coffee shop	Intermediate good
One share of Google stock	Neither
A new pick-up truck purchased by a consumer	Final good
A new home purchased by a family	Final good

5. Gross Domestic Product (GDP) is

- ☐ A. the quantity of all final goods and services produced in a country during a period of time.
- ☐ B. the value of all expenditures made by households and firms in a country during a period of time.
- ☒ C. the market value of all final goods and services produced in a country during a period of time.
- ☐ D. the amount of income generated from wages, rents, and profits in a country during a period of time.

6. Inventories are part of investment and therefore included in GDP because

- ☐ A. the value added by production goes into goods but not production and since goods start in inventory, that component needs to be computed.
- ☐ B. inventories represent the amount of factors of production firms have used to produce goods.
- ☐ C. the sale of shares of stock are not included in any other component, inventories allow GDP to capture this spending.
- ☒ D. firms produce goods, however sometimes the goods may be unsold at the time GDP is computed.

7. A simple economy produces four goods, apples, apple pies, cell phones, and software . All of the production of apples is used in the production of apple pies.

Use the information in the following table to compute GDP for the current year.

Product	Quantity	Price Per Unit
apples	115	\$0.75
apple pies	230	\$3.00
cell phones	700	\$650.00
software	1400	\$20.00

In this year, the nominal value of GDP in this economy is \$.

8. A recession occurs when an economy experiences a period in which there is

- ☒ A. a decrease in total production.
- ☐ B. an increase in total employment.
- ☐ C. a negative amount of net exports.
- ☐ D. a decrease in the inflation rate.

The business cycle exists because

- ☒ A. total production experiences periods of increases and periods of decreases.
- ☐ B. there are changes in the level of government expenditures, sometimes they increase and sometimes they decrease.
- ☐ C. the price level tends to increase over some periods and decrease over others.
- ☐ D. exports and imports do not always increase and decrease at the same rate which causes cyclical movements in net exports.

9. A farmer produces corn, which is valued at \$1.00. The corn is then sold to a miller, who turns it into cornmeal, which is valued at \$2.00. The miller then sells the cornmeal to a bakery, which uses it to make cornbread, valued at \$4.00. Finally, the bakery sells the cornbread to a grocery store, where it is sold to a consumer for \$8.00.

In the table below, show the value added at each stage of production, and the total value added to GDP.

Firm	Value Added
Farmer	\$ 1
Miller	\$ 1
Baker	\$ 2
Grocery Store	\$ 4
Total Value Added to GDP	\$ 8

10. Why does the size of a country's GDP matter?

The size of a country's GDP matters because

- ☒ A. GDP measures the level of income in the country during a period of time.
- ☐ B. GDP measures the total market value of concealed buying and selling transactions.
- ☐ C. GDP measures the total market value of goods and services produced in a country during the last ten years.
- ☐ D. GDP measures the total quantity of goods and services produced in a country during a period of time.

How does the size of a country's GDP affect the quality of life of the country's people?

- ☐ A. When GDP is high, production from the underground economy, and therefore the crime rate, is high.
- ☐ B. Generally, the more goods and services people have, the worse off they are.
- ☒ C. Generally, the more goods and services people have, the better off they are.
- ☐ D. There is an inverse relationship between GDP and quality of life.

11. Is GDP a perfect measure of economic well-being?

Choose the correct answer below.

- ☐ A. Yes, because it not only measures the value of goods and services produced but also takes into account their distribution.
- ☐ B. Yes, because in measures all types of production.
- ☐ C. No, its in an imperfect measure because it does not measure foreign production and the government sector.
- ☒ D. No, its in an imperfect measure because it does not measure household production and the underground economy.

Even if GDP included these types of production, why would it still be an imperfect measure of economic well-being?

- ☐ A. The value of leisure is not included in GDP.
- ☐ B. GDP is not adjusted for pollution and it does not account for unequal income distribution.
- ☐ C. GDP is not adjusted for crime or other social problems.
- ☒ D. All of the above.
- ☐ E. A and C only.

12. Which of the following are likely to increase the measured GDP⁵, and which are likely to reduce it?
- a. When the fraction of women working outside the home decreases, the measured GDP decreases .
 - b. When there is a sharp increase in the crime rate, the measured GDP may increase or decrease .
 - c. If lower tax rates cause some people to hide less of the income they earn, the measured GDP increases .

5: Definition

When Statistics Canada calculates GDP, it includes consumption expenditure, investment spending, government expenditure, and net exports. It does not include two types of production: production in the home and production in the underground economy.

13. Inventories are part of investment and therefore included in GDP because
- ☐ A. the sale of shares of stock are not included in any other component; inventories allow GDP to capture this spending.
 - ☒ B. firms produce goods, however sometimes the goods may be unsold at the time GDP is computed.
 - ☐ C. the value added by production goes into goods but not production, and since goods start in inventory, that component needs to be computed.
 - ☐ D. inventories represent the amount of factors of production firms have used to produce goods.
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14. Does inflation make nominal GDP a poor measure of the increase in total production from one year to the next?

Choose the correct answer below.

- ☐ A. No, nominal GDP is measured in quantity terms that are not influenced by inflation.
- ☒ B. Yes, when there is inflation, nominal GDP overstates the increase in total production.
- ☐ C. No, nominal GDP is evaluated at current prices and thus is not influenced by inflation.
- ☐ D. Yes, nominal GDP is evaluated at base-year prices that change due to inflation.

How does Statistics Canada deal with the problem inflation causes with nominal GDP?

- ☐ A. Statistics Canada measures GNP in base-year prices to fix the price level.
 - ☐ B. Statistics Canada separates price changes from quantity changes by using GDP deflator.
 - ☐ C. Statistics Canada fixes the quantity of goods and services produced to evaluate only an increase caused by change in price level.
 - ☒ D. Statistics Canada separates price changes from quantity changes by using chain-weighted prices.
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15. How is the GDP deflator calculated?

- ☒ A.
$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$
 - ☐ B.
$$\text{GDP deflator} = \frac{\text{Real GDP}}{\text{Nominal GDP}} \times 100$$
 - ☐ C.
$$\text{GDP deflator} = (\text{Nominal GDP} - \text{Real GDP}) \times 100$$
 - ☐ D.
$$\text{GDP deflator} = (\text{Nominal GDP} + \text{Real GDP}) \times 100$$
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16. A simple economy produces two goods, Pumpkin Pies and Video Games. Price and quantity data are as follows:

Production and Prices in Year 1 (Base year)		
Product	Quantity	Price Per Unit
Pumpkin Pies	110	\$1.00
Video Games	750	\$45.00

Production and Prices in Year 2		
Product	Quantity	Price Per Unit
Pumpkin Pies	137.50	\$1.50
Video Games	1,125.00	\$90.00

In Year 2, nominal GDP is equal to: \$, and real GDP is \$ (enter both responses rounded to the nearest penny).

17. Click on the graph in the window on the right and select *Multiple Time Series* to graph nominal GDP and real GDP for the years 1990-2006. For Y axis1 select nominal GDP, and for Y axis2 select real GDP. Roll your cursor over the plotted data to identify each series.

For this data, the base year for calculating GDP is 2000.

Use the graph to help determine which one of the following statements regarding nominal and real GDP is FALSE.

- ☒ A. When nominal GDP increases, real GDP falls.
- ☐ B. In 2000, nominal and real GDP were equal.
- ☐ C. In the 1990s, prices were, on average, lower than in 2000, so nominal GDP was lower than real GDP.
- ☐ D. After 2000, prices have been, on average, higher than in 2000, so nominal GDP is higher than real GDP.

Nominal GDP and Real GDP, 1990-2006

ERROR!!mygraph

Data source: Economic Report of the President

18. The production of LCD television sets in Australia increased by 20% this year as compared to last year. Those TV sets were no less advanced technologically than the sets sold in previous years. However, the value of the contribution LCD television sets made to Australia's nominal GDP during this year was the same as the previous year.

Which of the following, if true, would best explain this outcome?

- ☐ A. In the past year, the price of LCD TV sets in Australia has increased by a larger percentage than the percentage increase in the number of television sets produced.
- ☐ B. The contribution of competing products to this year's nominal GDP also remained unchanged.
- ☒ C. The price of LCD television sets declined in the past year.
- ☐ D. The size of the market for LCD televisions in Australia is smaller than the size of the market for LCD televisions in other countries.
- ☐ E. The increase in demand for LCD TV sets in Australia outweighed the increase in production this year.

19. Michael Burda of Humboldt University in Germany and Daniel Hamermesh of the University of Texas examined how workers in the United States who lost their jobs between 2003 and 2006 spent their time. They discovered that during the period when they were unemployed, the reduction in the number of hours of paid work by these workers was almost the same as the increase in the number of hours these workers devoted to household production.

Source: Michael Burda and Daniel S. Hamermesh, "Unemployment, Market Work, and Household Production," *Economic Letters*, Vol. 107, May 2010, pp. 131-133.

Based on Burda and Hamermesh's findings, what can we predict about total production in the economy – whether that production is included in the calculation of GDP or not – fell when these workers became unemployed?

- ☒ A. If the workers had been paying other people to perform the household activities prior to unemployment, then total production will fall.
- ☐ B. When workers lose their jobs, total production falls since workers are no longer earning wages.
- ☐ C. Since household production is not calculated in GDP, there is a decrease in total production that comes due to unemployment.
- ☐ D. If the workers had been paying other people to perform the household activities prior to unemployment, then total production will rise.

20. Is the value of a house built in 2010 and resold in 2022 included in the GDP⁶ of 2022? Briefly explain.

- ☐ A. Yes. The value of the house is included in gross fixed capital formation for 2022.
- ☐ B. Yes. The value of the house is included in final consumption expenditures for 2022.
- ☐ C. No. GDP for 2022 includes only the market value of final goods. A house resold in 2022 is an intermediate good.
- ☒ D. No. GDP for 2022 includes only production that occurs during 2022.

Would the services of a real estate agent who helped sell (or helped buy) the house be counted in GDP for 2022? Briefly explain.

- ☐ A. Yes. Real estate services are included in gross fixed capital formation.
- ☐ B. No. The value of the house is not included in GDP for 2022. Therefore, real estate services are not included either.
- ☐ C. No. GDP for 2022 includes only the market value of final goods. Real estate services are not final goods.
- ☒ D. Yes. GDP for 2022 includes the market value of final goods and services. This includes real estate services.

6: Definition

Gross domestic product (GDP) is the market value of all final goods and services produced in a country during a period of time, typically one year.

21. Suppose the information in the following table is for a simple economy that produces only the following four goods: textbooks, hamburgers, shirts, and cotton. Further, assume that all of the cotton is used to produce shirts.

Product	2009		2018		2019	
	Quantity	Price	Quantity	Price	Quantity	Price
Textbooks	90	\$50.00	100	\$60.00	100	\$65.00
Hamburgers	80	2.50	110	2.50	125	2.75
Shirts	50	30.00	50	25.00	65	25.00
Cotton	120	0.90	1,200	0.75	1,300	0.80

- a. If the base year is the year 2009, then real GDP⁷ for 2018 equals \$. (Round your response to the nearest cent.)

The real GDP for 2019 equals \$. (Round your response to the nearest cent.)

- b. The (annual) growth rate of real GDP during 2019 is %. (Round your response to two decimal places.)

7: Definition

Real GDP is the value of final goods and services evaluated at base-year prices.

22. Use the data in the following table to calculate the GDP deflator⁸ for each year (values are in billions of dollars). (Round your responses to one decimal place. Remember to multiply by 100.)

Year	Nominal GDP	Real GDP	GDP Deflator
2014	\$16,152	\$15,372	105.1
2015	16,673	15,592	106.9
2016	17,397	15,965	109.0
2017	18,041	16,390	110.1
2018	18,571	16,679	111.3

Which year from 2015 to 2018 saw the largest percentage increase in the price level, as measured by changes in the GDP deflator?

The largest percentage increase in the price level was in year .

8: Definition

The GDP deflator is a measure of the price level, calculated by dividing nominal GDP by real GDP and multiplying by 100.